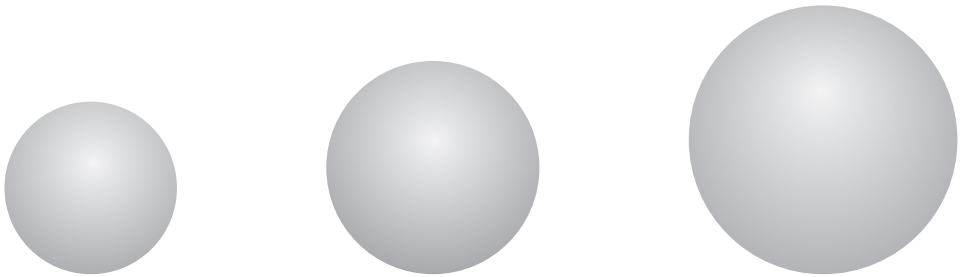
**12** Here are three spheres.



**P** **Q** **R**

The volume of sphere **Q** is 50% more than the volume of sphere **P**.
The volume of sphere **R** is 50% more than the volume of sphere **Q**.

Find the volume of sphere **P** as a fraction of the volume of sphere **R**.

A 50% increase means multiply by 1.5.

| Percent | Multiplier |
|---|---|
| 100 | 1 |
| 150 | 1.5 |

×1.5 ↓ ×1.5

**1 mark**
Q is 1.5 times P.

Compare the volumes P : Q : R.

| P | Q | R |
|---|---|---|
| 1 | 1.5 | 2.25 |
| 4 | 6 | 9 |

×4 ↓ ×4

**1 mark**
R = 2.25 times P.

**1 mark**
P/R = 4/9.

4/9

**(Total for Question 12 is 3 marks)**